

Push your Terraform code from an Amazon Linux EC2 instance to GitHub manually.

Steps to Push Terraform Code to GitHub from EC2

1. Install Git (if not already installed)

```
sudo yum install git -y
```

2. Configure Git

Set your Git username and email:

```
git config --global user.name "Your Name"  
git config --global user.email "your.email@example.com"
```

3. Initialize a Git Repository

Navigate to your Terraform code directory:

```
cd /path/to/your/terraform/code
```

Initialize the Git repository:

```
git init
```

4. Add Your Files

```
git add .  
git commit -m "Initial commit of Terraform code"
```

5. Connect to GitHub

You have two options to authenticate:

Option A: Using HTTPS with Personal Access Token

1. Go to GitHub and create a new repository.
2. Copy the HTTPS URL of the repo.
3. Push your code:

```
git remote add origin https://github.com/your-username/your-repo.git
git push -u origin master
```

Git will prompt for your GitHub username and **personal access token** (not your password).

Option B: Using SSH (Recommended for EC2)

1. Generate an SSH key:

```
ssh-keygen -t rsa -b 4096 -C "your.email@example.com"
```

Press Enter to accept defaults.

2. Add the public key to GitHub:

```
cat ~/.ssh/id_rsa.pub
```

Copy the output and add it to GitHub:

- Go to GitHub > Settings > SSH and GPG keys > New SSH key
- Paste the key and save

3. Test the connection:

```
ssh -T git@github.com
```

4. Add the remote and push:

```
git remote add origin git@github.com:your-username/your-repo.git
git push -u origin master
```

5. Update Git Remote to Use SSH

Run this command inside your Git repository:

```
git remote set-url origin git@github.com:<repo\_name>.git
```

You can verify the change with:

```
git remote -v
```

6. Now Try Pushing Again

```
git push origin main
```